Recent advances in berry supplementation and age-related cognitive decline.

PURPOSE OF REVIEW: To summarize recent findings and current concepts in the beneficial effects of berry consumption on brain function during aging. RECENT FINDINGS: Berryfruit supplementation has continued to demonstrate efficacy in reversing age-related cognitive decline in animal studies. In terms of the mechanisms behind the effects of berries on the central nervous system, recent studies have demonstrated the bioavailability of berry polyphenols in several animal models. These studies have revealed that flavonoids and polyphenols from berries do accumulate in the brain following long-term consumption. Finally, several compelling studies have revealed that berries can influence cell-signaling cascades both in vivo and in cell culture systems. These studies underscore the developing theory that berries and antioxidant-rich foods may be acting as more than just oxygen radical neutralizers in the aging central nervous system. SUMMARY: Antioxidant-rich berries consumed in the diet can positively impact learning and memory in the aged animal. This effect on cognition is thought to be due to the direct interaction of berry polyphenols with aging neurons, reducing the impact of stress-related cellular signals and increasing the capacity of neurons to maintain proper functioning during aging.